

Serving = four concerns, not a stack

orthogonal decisions - they interact, but none sits on top of another

**Request stream
(text → tokens)**

THE FOUR SERVING CONCERNS

REUSE

prefix cache (RadixAttention / APC):
reuse KV of shared context

re-prefill FLOPs & memory skipped

REQUEST SCHEDULING

continuous batching: admit &
evict at every decode step

scheduling ↔ latency · utilization

RESOURCE SPECIALISATION

P/D split: prefill pool (compute) +
decode pool (bandwidth)

two regimes → two pools

all four in parallel

STATE MANAGEMENT

KV cache in fixed pages
(PagedAttention page table)

memory capacity → concurrency

Output tokens